Product Requirements Document (PRD)

Project: EduLLM — A Language Model Trained on Engineering Educational Content

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1. Executive Summary

• EduLLM is an AI-powered institutional assistant built using a lightweight, domain-specific large language model (LLM) tailored for engineering education.
• It provides intelligent summarization, contextual question answering, and document-driven insights through a visual analytics dashboard.
• The project combines transformer-based architectures with parameter-efficient fine-tuning to make it computationally efficient and domain-aware.
• The system aims to support students, faculty, and administrators with a scalable and easily deployable solution.

2. Problem Statement

• Current LLMs (like GPT-3 or GPT-4) are too generalized and resource-intensive for educational use in engineering institutions.
• Students and faculty struggle with lack of domain-specific assistance that understands technical content.
• Institutions face difficulty in extracting insights from academic data and documents without complex manual analysis.
• This results in inefficiency, reduced accessibility, and slower decision-making in academic processes.

3. Proposed Solution

• Develop EduLLM — a fine-tuned, domain-specific model trained exclusively on engineering educational data.
• Include features such as:
  - Context-aware Q&A and document summarization.
  - Role-based dashboards for students, faculty, and administrators.
  - Parameter-efficient architecture for lower hardware dependency.
  - Modular, scalable design with visual analytics.
• EduLLM acts as both a learning assistant and decision-support system for institutions.

4.Requirements:

- ### User Stories

  • **Student** – As a student, I want to ask questions about my coursework so that I can get accurate, syllabus-based answers.
  Acceptance Criteria:
  - Contextual, domain-relevant answers
  - Response time under 3 seconds
  - Confidence score of at least 85%

  • **Faculty** – As a faculty member, I want to summarize academic papers so I can review key findings quickly.
  Acceptance Criteria:
  - Summaries with less than 10% information loss
  - Generates results under 10 seconds
  - Supports PDF and DOCX uploads

  • **Administrator** – As an administrator, I want a visual dashboard showing performance and trends to make better decisions.
  Acceptance Criteria:
  - Dashboard loads in under 5 seconds
  - Real-time updates and export options available

  • **Researcher** – As a researcher, I want to fine-tune the LLM on new data to evaluate emerging topics.
  Acceptance Criteria:
  - Dataset upload and training monitoring supported
  - Real-time metrics visible

- ### Functional Requirements:

  • **FR-1:** Support document uploads (PDF, DOCX, TXT)
  • **FR-2:** Provide summarization (extractive + abstractive)
  • **FR-3:** Enable context-aware question answering
  • **FR-4:** Offer dashboards with analytics and insights
  • **FR-5:** Include secure login with role-based access control
  • **FR-6:** Allow researchers to fine-tune EduLLM with institutional datasets

- ### Non-Functional Requirements:

  • **Performance:** Inference under 3 seconds, summarization under 10 seconds
  • **Scalability:** Handles over 100 concurrent users
  • **Security:** Role-based data access with encryption
  • **Usability:** Clean, intuitive interface for all roles
  • **Maintainability:** Modular architecture for easy updates
  • **Compatibility:** Works on all modern browsers and operating systems

5. Planned Milestones

• **Stage 1:** Planning, data collection, and preprocessing
• **Stage 2:** Core AI model setup and knowledge retrieval
• **Stage 3:** Application development, access control, and testing
• **Stage 4:** Final testing, deployment, and publication

7. Success Metrics / KPIs:

- Model Accuracy & Relevance
- Performance & Responsiveness
- User Engagement & Satisfaction
- System Scalability & Reliability
- Research & Training Effectiveness
- Security & Compliance

8. Empirical Validation:

- Automated Logging & Monitoring
- User Surveys & Feedback
- Usage Analytics & Adoption Rates
- Performance Benchmarks
- Security & Compliance Audits
- Training & Fine-Tuning Metrics

6. Sign-Off

- **Teams Involved:**
  • AI/ML Development Team
  • Backend & Frontend Developers
  • Faculty Advisor / Principal Investigator

- **Sign-off Required From:**
  • Project Lead
  • Department Head
  • Institutional Research Committee